📅 Day 5 — Autoregressive Models

1. Sequential Data

Many real-world datasets are **sequences**, meaning the **order of elements matters**.

Examples:

- Natural language text
- Speech signals
- Time-series data
- Video frames

Example sentence:

```
I love machine learning
```

If we change the order:

```
learning machine love I
```

The meaning changes completely.

Therefore, models must **understand the order of data**.

2. Sequential Models

A **Sequential Model** is a model designed to **process ordered data** where each element depends on previous elements.

These models analyze sequences and learn patterns within them.

Examples of Sequential Models

- RNN (Recurrent Neural Network)
- LSTM (Long Short-Term Memory)
- GRU (Gated Recurrent Unit)
- Transformer models

Example Task

Input sentence:

```code
The movie was amazing
```

Output:

```code
Positive Sentiment
```

Here the model **reads the sequence and produces a prediction**.

Sequential models are used for:

- sentiment analysis
- speech recognition
- machine translation
- time-series prediction

In sequential models:

1. Data enters as a sequence
2. The model processes information step-by-step
3. A prediction is produced at the end

These models mainly **understand sequences rather than generate them**.

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3. Limitation of Sequential Models

Sequential models are good at **analyzing sequences**, but many applications require **generating sequences**.

Example prompt:

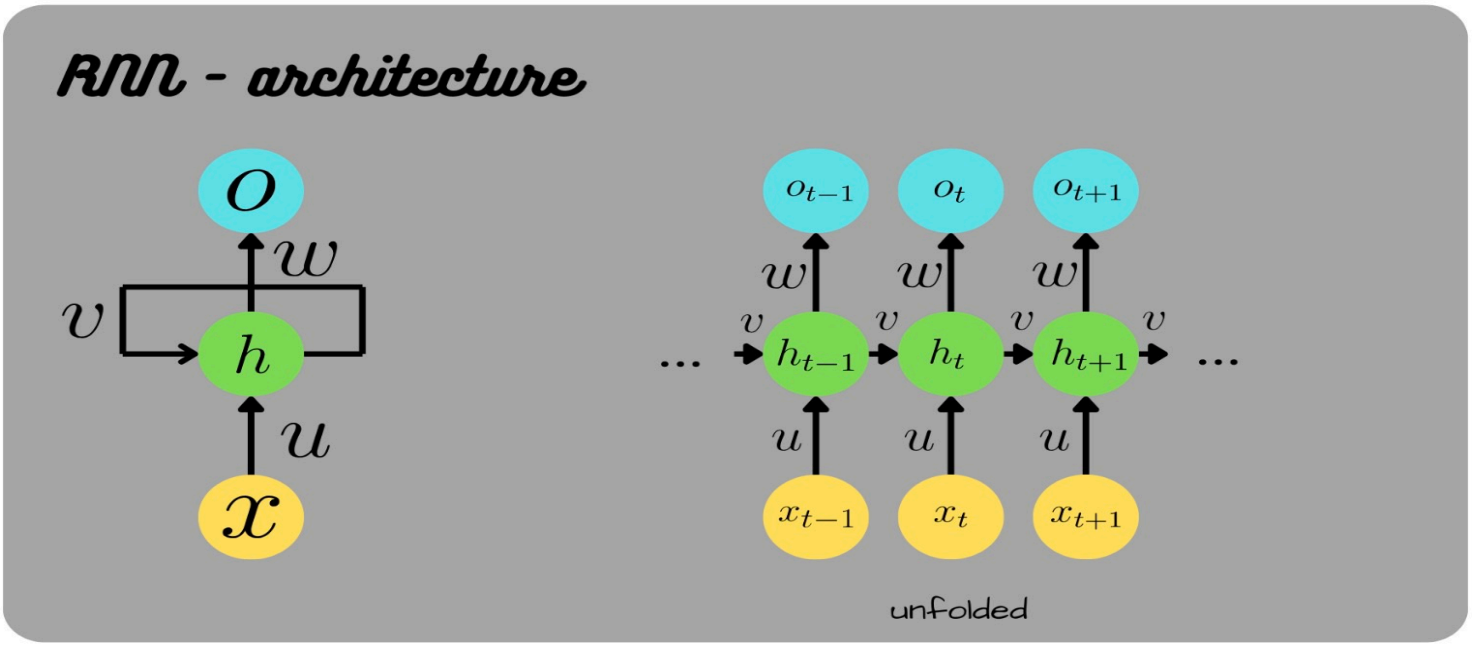
```
code
Artificial Intelligence will
```

We want the model to generate:

```
code
Artificial Intelligence will transform industries
```

This requires predicting **the next word repeatedly**.

To solve this problem, we use **Autoregressive Models**.



4. Autoregressive Models

An **Autoregressive Model** is a generative model that **predicts the next element in a sequence using previous elements**.

The model generates output **step-by-step**.

Example generation:

```code
AI
AI will
AI will change
AI will change the
AI will change the world
```

Each new word depends on **previous words**.

Mathematical Idea

Autoregressive models follow the **chain rule of probability**.

$$P(x_1, x_2, x_3, ..., x_n) = P(x_1) \times P(x_2|x_1) \times P(x_3|x_1, x_2) \times ... \times P(x_n|x_1...x_{n-1})$$

Meaning:

The probability of a sequence is computed using **conditional probabilities of previous tokens**.

Autoregressive generation process:

1. Input prompt is given
2. Model predicts the next token
3. The predicted token is appended to the sequence
4. Prediction continues until completion

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5. Training Process

Autoregressive models are trained using **next-token prediction**.

Example sentence:

```
Deep learning models are powerful
```

Training pairs:

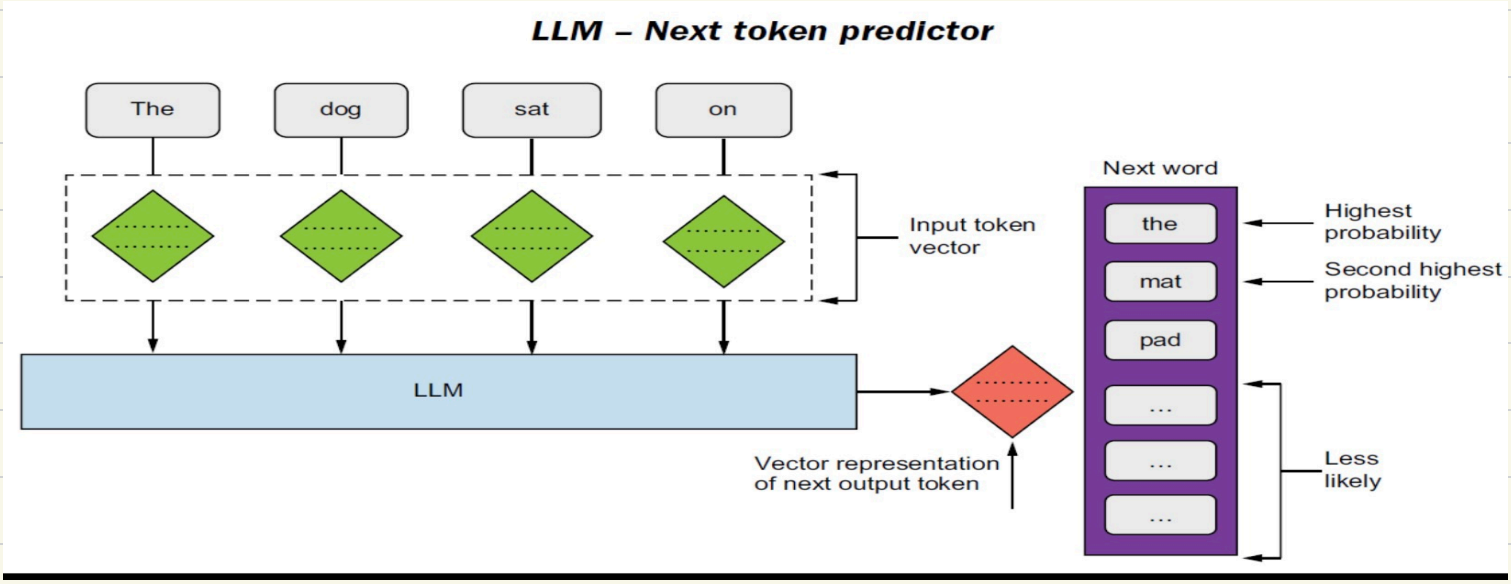
Input	Target
Deep	learning
Deep learning	models
Deep learning models	are
Deep learning models are	powerful

The model learns:

```
P(next_token | previous_tokens)
```

Loss function used:

Cross-Entropy Loss



6. Architecture Used

Modern autoregressive systems use **Transformer Decoder architecture**.

Important components:

Tokenization

Text → tokens

```code
Machine learning is powerful
```

Tokens:

```code
[Machine, learning, is, powerful]
```

Embeddings

Tokens are converted into **numerical vectors**.

Positional Encoding

Adds position information to maintain sequence order.

Masked Self-Attention

The model **cannot see future tokens** during prediction.

Example:

```code
AI will transform industries
```

When predicting **transform**, the model sees:

```code
AI will
```

7. Relationship Between Sequential and Autoregressive Models

Sequential models process sequences, while autoregressive models generate sequences.

Relationship:

```
code                                                    Copy

Sequential Models
        │
        ├── Sequence Analysis
        ├── Sequence Prediction
        └── Autoregressive Models
                │
                └── Sequence Generation
```

So:

Autoregressive models are **a special form of sequential modeling used for generation tasks**.

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8. Comparison

Feature	Sequential Model	Autoregressive Model
Purpose	Understand sequences	Generate sequences
Output	Label/value	Generated tokens
Prediction style	Single prediction	Step-by-step
Example task	Sentiment analysis	Text generation
Architecture	RNN, LSTM, Transformer	Transformer decoder

9. Latest Real-World Examples (2025–2026)

Modern generative AI systems rely heavily on autoregressive models.

Examples include:

- ChatGPT
- GPT-4
- GPT-5

These systems generate responses **token by token using autoregressive decoding**.

Example prompt:

```
code
Explain generative AI
```

The model generates the response **word by word until completion**.

Other applications include:

- Code generation using GitHub Copilot
- conversational AI
- automated content generation
- AI assistants